

A dramatic landscape photograph featuring a large, rugged mountain peak with a flat top, partially shrouded in thick, low-hanging clouds. The sky is filled with soft, golden-brown clouds, suggesting a sunrise or sunset. The foreground shows dark, forested slopes. The text "Virtual Lab Setup" is centered over the image in a white, serif font.

Virtual Lab Setup

Step 1: Install Virtualization Software

Ensure VirtualBox is installed. If not, use the following command:

```
sudo pacman -S virtualbox
```

Step 2: Create a New Virtual Machine

- Open VirtualBox and click "New."
- Name the VM, select "Linux" for type, and "Ubuntu (64-bit)" for version.
- Allocate appropriate memory and create a virtual hard disk.

Step 3: Install Ubuntu

- Attach the 'boonter.iso' to the VM's virtual CD/DVD drive.
- Start the VM and follow the installation prompts to install Ubuntu.

Step 4: Configure Network Settings

- **NAT:** For internet access.
- **Internal Network:** For VM to VM communications within the isolated lab environment.

Step 5: Install Necessary Tools

Install tools for network testing and attacks and the attacking machine:

```
sudo apt update  
sudo apt install tcpdump ettercap-text-only
```

Step 6: Basic Network Testing

- Perform network tests like pinging and capturing packets with tcpdump.
- Analyze the traffic using Wireshark to ensure proper setup.

Step 7: Security Practices

- Keep the VMs updated and secure.
- Enable only necessary network interfaces and services.
- Regularly snapshot the VMs for easy restoration.

Network Setup for TCP Session Hijacking Lab

Choosing an IP Scheme and Assigning IPs

- Use the private IP range `10.0.2.x` with a subnet mask of `255.255.255.0`.
- Assign static IPs:
 - Attacker VM: `10.0.2.10`
 - Victim VM: `10.0.2.20`
 - Server VM: `10.0.2.30`

Setting Up the Network in VirtualBox

- For each VM in VirtualBox:
 - Go to 'Settings' -> 'Network'
 - Enable a network adapter and set it to 'Internal Network'.
 - Use 'LabNet' for the network name to ensure all VMs can communicate.

Configuring Static IP Addresses on Ubuntu VMs Without Gateway

- Edit the Netplan configuration file `/etc/netplan/01-netcfg.yaml` (or similar):

```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 10.0.2.10/24 # Use the appropriate IP for each VM
```

- Adjust the IP for each VM accordingly.
- Apply changes with `sudo netplan apply`.

Configuring Static IP Addresses on Ubuntu VMs With Gateway

- Edit the Netplan configuration file `/etc/netplan/01-netcfg.yaml` (or similar):

```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      dhcp4: no
      addresses: [10.0.2.10/24]
      gateway4: 10.0.2.1
      nameservers:
        addresses: [8.8.8.8]
```

- Adjust the IP for each VM accordingly.
- Apply changes with `sudo netplan apply`.

Testing Connectivity

- Test network connectivity using `ping` from each VM:

```
ping 10.0.2.20 # From VM with IP 10.0.2.10
```